

AYUSH JUVEKAR

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Computer Vision / Edge AI Engineer

PROFILE

Computer vision engineer focused on video analysis, edge inference, object detection workflows, and camera/sensor-backed research prototypes.

EDUCATION

Michigan Technological University, Houghton, MI
M.S. Computer Science

CGPA: 3.9/4.0
Graduated Apr 24, 2026

Pune Institute of Computer Technology, Pune, India
B.E. Computer Engineering

GPA: 8.14/10
Jun 2024

EXPERIENCE

Freelance Software Developer | Coherence Academy
Pune, India

Aug 2026 - Present

- Building Coherence Fee Desk, a Next.js platform replacing the academy's spreadsheet-based admissions, fee collection, and reporting.
- Separately run the software for Coherence-backed events (Campus Connect, Mission 95): registration, UPI payments, Sheets sync.

Teaching Assistant - Physics Department | Michigan Technological University
Houghton, MI

Sep 2024 - Apr 24, 2026

- Supervised and graded physics labs for 100+ students across four semesters, guiding experiments, explaining concepts, and providing clear feedback across five lab sections per semester.

SELECTED PROJECTS

Crowd Anomaly Detection Prototype | *Python, TensorFlow, OpenCV, TensorFlow Lite, Jetson Nano*

- Led model research and data collection for a BE final-year team project on edge crowd-anomaly detection, recording training videos and evaluating CNN/TensorFlow approaches on 10K+ frames.
- Optimized the prototype with TensorFlow Lite for Jetson Nano demonstration, reaching about 92% test accuracy in controlled demo runs.

Particle Vibration Analysis | *MATLAB, Computer Vision, Object Detection, ZED 2 Camera*

- Built MATLAB computer-vision scripts for particle vibration analysis from ZED 2 camera video, using object detection workflows to track particles in test recordings.
- Delivered analysis outputs for a professor's downstream PhD research, validating scripts on sample particle videos before larger research use.

IoT Air Quality Monitoring System | *Arduino Uno, C++, MQTT, Zigbee, Node-RED* | [IEEE](#)

- Co-built an IEEE-published IoT air-quality prototype for CO/gas monitoring, using Arduino Uno/C++, MQTT/Zigbee communication, and Node-RED flows to automate air-filter behavior from live sensor readings.
- Owned the Node-RED automation and conference presentation for the two-person project, contributing to the paper/report accepted at ESCI 2023.

TECHNICAL SKILLS

Languages Python, C++, MATLAB

Computer Vision OpenCV, MATLAB Computer Vision workflows, image/video processing, object detection

ML / Edge AI TensorFlow, TensorFlow Lite, CNNs, Jetson Nano

Sensors / Cameras ZED 2 camera video, Arduino sensor data

Data Workflows Dataset preparation, model evaluation, experiment documentation

Tools Jupyter Notebook, Git, Linux

PUBLICATION

A. Ghadge, A. Juvekar et al., "Carbon Monoxide Concentration Monitoring System for Automating Air Filters," *ESCI 2023*. [IEEE](#)